CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 6

DIVISION OF TWO 8-BIT NUMBERS

AIM: To write and execute an 8085 assembly language program to divide two 8-bit numbers and store the result in memory.

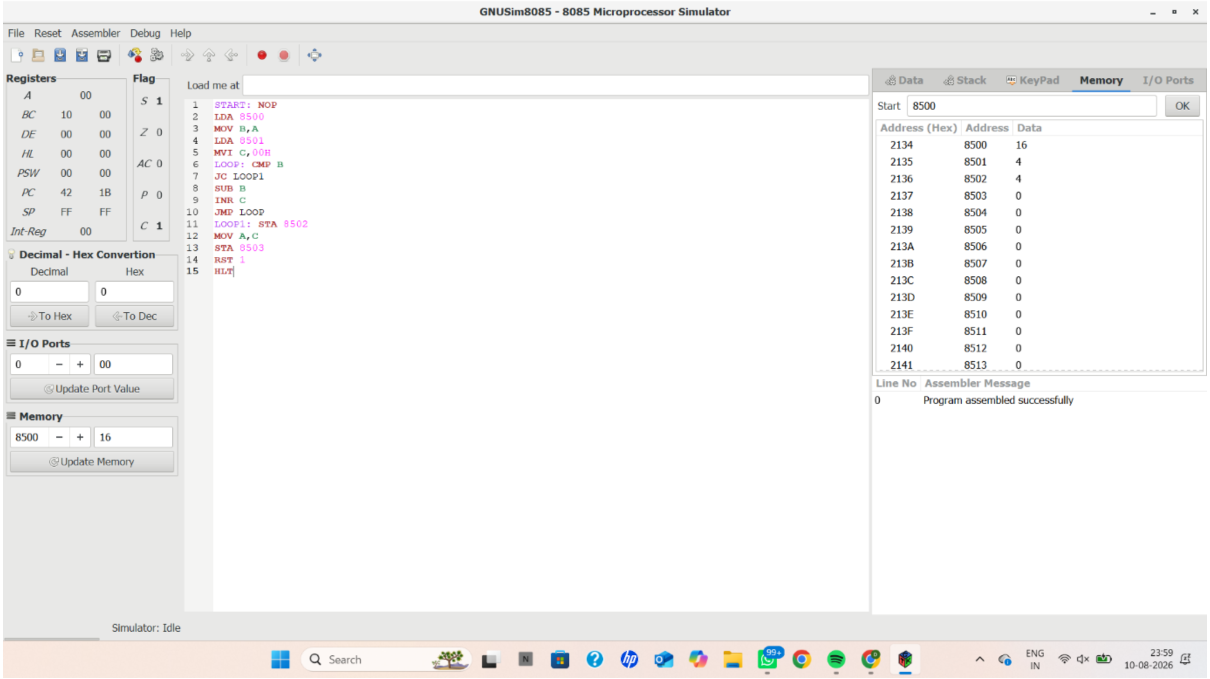
ALGORITHM

1. Load the dividend into register A.
2. Load the divisor into register B.
3. Initialize register C to 00H.
4. Compare the dividend with the divisor.
5. If the dividend is smaller, go to the end.
6. Subtract the divisor from the dividend.
7. Increment register C.
8. Repeat steps 4–7 until the dividend is smaller than the divisor.
9. Store the remainder and quotient in memory.
10. Stop.

CODE:

```
MVI A,14H

MVI B,04H

MVI C,00H

LOOP: CMP B

JC END

SUB B

INR C

JMP LOOP

END: STA 8050H

MOV A,C

STA 8051H

HLT
```

OUTPUT:



GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
Flag
Load me at
START: NOP
LDA 8500
MOV B,A
LDA 8501
MVI C,00H
LOOP: CMP B
JC LOOP1
SUB B
INR C
JMP LOOP
LOOP1: STA 8502
MOV A,C
STA 8503
RST 1
HLT
Data Stack KeyPad Memory I/O Ports
Start 8500 OK
Address (Hex) Address Data
2134 8500 16
2135 8501 4
2136 8502 4
2137 8503 0
2138 8504 0
2139 8505 0
213A 8506 0
213B 8507 0
213C 8508 0
213D 8509 0
213E 8510 0
213F 8511 0
2140 8512 0
2141 8513 0
Line No Assembler Message
0 Program assembled successfully
Decimal - Hex Convertion
To Hex
To Dec
I/O Ports
Update Port Value
Memory
Update Memory
Simulator: Idle